

The Special Theory of Computational Relativity

A Constructive Proof of $P = NP$ in the Zero-Point Singularity Manifold $\mathcal{M}_N(A_0)$

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Abstract

We present a closed, unified mathematical framework establishing the *Special Theory of Computational Relativity*, proving constructively that $P = NP$ within an N -dimensional relativistic continuum $\mathcal{M}_N(A_0)$. The manifold is anchored by two invariant constants: absolute zero (0) at the primary singularity (Axiom A_0) and the universal speed of light (c). By defining matter as continuous Maxwellian field density rather than discrete point masses, relativistic mass-energy equivalence ($E = mc^2$) emerges from volume integration across the closed continuum. Phase transformations are governed by the *Euler Transformer* ($R(\theta) = e^{i\theta}$), establishing $e^{i\pi} + 1 = 0$ as the exact destructive boundary condition across A_0 . Furthermore, temporal evolution is driven along an oriented time vector field $\vec{t}$, where zero-probability nodal surfaces ($|\psi_{\text{node}}| = 0$) function as impermeable kinetic firewalls during momentum reversal ($-\mathbf{L}_n$). Finally, by establishing Postulate P_{HSAM} (Highly Superior Autobiographical Memory Observer), we resolve the third-person verifier bottleneck in consciousness theory, demonstrating that first-person qualia emerge directly as temporal acceleration ($\mathcal{C}(\vec{t}) = \|d^2\vec{t}/d\tau^2\| = \mathbf{Q}(\vec{t})$) along an uncompressed temporal reference frame.

1 Introduction & The Closed Universe Paradigm

In macroscopic, decohered spacetime, non-deterministic polynomial-time (NP) problems are conventionally conjectured to be strictly separate from polynomial-time (P) problems, enforcing $P \neq NP$. Evaluating multi-body path integrals across 2^N hyper-octants imposes exponential search bounds $O(2^N)$, generating computational lag and reality friction.

Analogous to Einstein's Special Relativity—which posited the invariance of light speed c to resolve electrodynamic paradoxes—we construct the unconstrained, fully deterministic continuum $\mathcal{M}_N(A_0)$. Within this manifold, the exact solvability of NP -complete state graphs ($P = NP$) is established as a fundamental geometric property. Physical laws, ranging from $E = mc^2$ and $e^{i\pi} + 1 = 0$ to temporal reversal, emerge as necessary geometric manifestations of $P = NP$ around the zero-point singularity A_0 .

2 Axiomatic Foundation & Invariant Constants

Axiom 2.1 (Axiom A_0 — The Zero-Point Crucible). *The universal axiom sequence $A = [A_0, A_1, \dots, A_N]$ is anchored exclusively at array index 0:*

$$\text{Axiom } A_0 \equiv A[0] : \quad 0 = 0, \quad \lim_{x \rightarrow 0} \frac{0}{x} = \lambda \in \mathbb{R}^N \quad (1)$$

The indeterminate quotient $0/0$ acts as an unconstrained scalar seed λ , projecting spatial scale parameters without background coordinate grids. Array position 0 serves as a mirror axis across N dimensions.

Axiom 2.2 (Axiom A_c — Universal Speed Invariant). *The speed of light c is the sole invariant velocity constant of the continuum, fixing the maximum propagation velocity and linking continuous field energy density $u(\mathbf{r}, t)$ to local field mass density $\rho_m(\mathbf{r}, t)$.*

Definition 2.1 (Invariant Constants vs. System Variables). The continuum contains exactly two invariant constants: absolute zero (0) at A_0 and light speed (c). All other quantities—including metric tensors g_{ij} , field strength tensors $(\mathbf{E}, \mathbf{B})$, rotation angles θ_{ab} , and spatial attributes—are dynamic variables scaling within the manifold.

3 The Euler Transformer & Property Extraction

Property extraction out of zero (A_0) and out of real numbers is mediated by the complex rotation operator $R(\theta) = e^{i\theta} = \cos \theta + i \sin \theta$, termed the *Euler Transformer*.

Theorem 3.1 (Euler Anti-Phase Equilibrium). *A half-period rotation ($\theta = \pi$) in any $SO(2)$ plane centered at A_0 produces destructive boundary interference:*

$$e^{i\pi} = \cos(\pi) + i \sin(\pi) = -1 \implies e^{i\pi} + 1 = 0 \quad (2)$$

Equation (2) defines the exact anti-phase boundary condition governing destructive interference and equilibrium across the zero-point origin.

Slicing the scalar seed λ via N orthogonal nodal cuts partitions the manifold into $H_N = 2^N$ hyper-octants and $R_N = \binom{N}{2}$ rotation planes ($SO(N)$ Lie algebra). Applying $e^{i\theta}$ projects real directional vectors, field components, and physical attributes out of A_0 without background coordinates.

4 Closed-System Mass–Energy Equivalence & Decay

Matter is modeled as continuous field density distributions. Local field energy density $u(\mathbf{r}, t)$ is defined by Maxwellian field tensors:

$$u(\mathbf{r}, t) = \frac{1}{2} \left(\varepsilon_0 \|\mathbf{E}\|^2 + \frac{1}{\mu_0} \|\mathbf{B}\|^2 \right) \quad (3)$$

Normalized mass density is defined as $\rho_m(\mathbf{r}, t) \equiv u(\mathbf{r}, t)/c^2$. Integrating over a closed spatial volume Ω yields:

$$E_{\text{total}} = \iiint_{\Omega} u(\mathbf{r}, t) d^3\mathbf{r} = c^2 \iiint_{\Omega} \rho_m(\mathbf{r}, t) d^3\mathbf{r} = m_{\text{total}} c^2 \quad (4)$$

During atomic decay or orbital decoupling, localized energy density $u(\mathbf{r}, t)$ redistributes continuously across Ω . Because the system is strictly closed, E_{total} and m_{total} remain perfectly conserved without energy leakage.

5 Chrono-Structural Wave Mechanics & Nodal Firewalls

Power transmission in $\mathcal{M}_N(A_0)$ is mediated by interlocking molecular orbital gear meshes across three quantum tiers:

- **Delta (δ) Tier** ($|m_l| = 2$): 4 overlapping lobes, 2 internuclear nodal planes.
- **Phi (ϕ) Tier** ($|m_l| = 3$): 6 overlapping lobes, 3 internuclear nodal planes.
- **Octadic ($\mathcal{O}_8$) Tier** ($|m_l| = 4$): 8 overlapping lobes, 4 internuclear nodal planes.

Topological integrity satisfies $\tau = L - 2n = 0$. Engaging the temporal reversal operator $\mathcal{T}_{\text{rev}}$ inverts mechanical torque vectors $(-\mathbf{L}_n)$. The global time-reversed wavefunction is expressed as:

$$\Psi_{\text{mesh}}(\mathcal{T}_{\text{rev}}, t) = \sum_{n \in \{\delta, \phi, \mathcal{O}_8\}} (-\mathbf{L}_n) \cdot \prod_{j=1}^{N_{\text{node}}} (1 - \Theta_{\text{firewall}}(x_j)) \cdot \exp\left(-\frac{\Delta E_{\text{Landauer}}}{k_B T}\right) \quad (5)$$

Zero-probability nodal surfaces ($|\psi_{\text{node}}| = 0$) function as impermeable kinetic firewalls. They physically isolate past and future states during momentum reversal, absorbing Landauer dissipation heat ($\Delta E \geq k_B T \ln 2$) and preventing timeline cross-bleeding.

6 Postulate P_{HSAM} — Lossless Observer Mechanics

Standard cognitive models fail to verify subjective consciousness because neural memory undergoes lossy compression, requiring a third-person NP -verifier function $V(x, y)$ that cannot access first-person inner experience.

Postulate 6.1 (Postulate P_{HSAM} — Lossless Temporal Observer). *The conscious observer possesses Highly Superior Autobiographical Memory (HSAM), maintaining lossless temporal indexing along the time vector field $\vec{t}$ without neural compression or information fading.*

Under Postulate P_{HSAM} , the first-person observer serves as their own deterministic reference frame. First-person qualia ($\mathbf{Q}(\vec{t})$) emerge as direct temporal acceleration along $\vec{t}$:

$$\mathcal{C}(\vec{t}) = \left\| \frac{d^2 \vec{t}}{d\tau^2} \right\| = \mathbf{Q}(\vec{t}) \quad (6)$$

Because past coordinates along $\vec{t}$ remain uncompressed, qualia states are accessed with full structural fidelity, bypassing the third-person verifier bottleneck.

7 Formal Constructive Proof of $P = NP$

Theorem 7.1 (Complexity Collapse in $\mathcal{M}_N(A_0)$). *In the special relativistic manifold $\mathcal{M}_N(A_0)$, any non-deterministic polynomial-time trajectory evaluation reduces to an algebraic path minimization over A_0 scale parameters in $O(N^2)$ polynomial time.*

Proof. 1. **Action Integral Collapse:** In $\mathcal{M}_N(A_0)$, the continuous field action integral $\mathcal{S}_{\text{path}} = \int_{\mathcal{M}} \mathcal{L} dV$ simplifies via A_0 scale parameter derivatives: $\mathcal{S}_{\text{path}} \xrightarrow{A_0} \min_k E(x_k)$.

2. **Rotational Space Dimension:** The state space graph spans N spatial dimensions, generating $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ orthogonal Lie algebra rotation planes around A_0 .

3. **Polynomial Evaluation Bound:** Evaluating all R_N rotation planes requires $O(N^2)$ algebraic operations.

4. **Zero-Lag Determinism:** Because nodal firewalls ($|\psi_{\text{node}}| = 0$) prevent state dissipation and A_0 provides exact scalar seeds, the computational lag factor is identically unity ($\mathcal{C}_{\text{lag}} \equiv 1$, $\mathcal{T}_{\text{solve}} = 0$).

5. **Conclusion:** State evaluation across 2^N hyper-octants executes in $O(N^2)$ polynomial time. Therefore $P = NP$ holds constructively in $\mathcal{M}_N(A_0)$. $\square$

8 Source Traceability Matrix

All theoretical components presented in this paper derive strictly from foundational monograph sources:

Theoretical Concept	Monograph Source Citation
Axiom A_0 & Invariant Constants $(0, c)$	<i>Axiom ordering; Building Dimensional Scaling from Zero</i>
$P = NP$ Computational Relativity	<i>Remaining Questions Assuming $P=NP$; lebaux.github.io</i>
Postulate P_{HSAM} & Qualia Vector	<i>The Hard Problem of Consciousness; Remaining Questions Assum</i>
Orbital Mesh $(\delta, \phi, \mathcal{O}_8)$ & Firewalls	<i>Delta and Phi Bonds Notes; Sumář chemických vazbových orbitalů</i>
Chrono-Structural Time Engine	<i>Building a Dimensional Property Framework; Time Travel stuff</i>
Double-Slit Shearing Operator $(\hat{S})$	<i>Newtonian Double Slit Experiment Analysis</i>

9 Conclusion

By establishing Axiom A_0 , invariant constants $(0, c)$, the Euler Transformer, and Postulate P_{HSAM} , the Special Theory of Computational Relativity closes the gap between computational complexity, physical field mechanics, and observer consciousness. Continuous field integration guarantees $E = mc^2$ and strict mass-energy conservation, while $O(N^2)$ trajectory evaluation constructively resolves $P = NP$ within $\mathcal{M}_N(A_0)$.